A REPORT ON

IOT-ENABLED SOLAR-ASSISTED HIGH-EFFICIENCY BATTERY CHARGING USING SUPERCAPACITOR ENERGY BUFFERING

BY

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ABSTRACT

The rapid adoption of electric vehicles and rooftop solar generation has increased the need for charging systems that can deliver power efficiently while remaining observable and controllable. Conventional residential charging draws entirely from the grid at a rate limited by the household connection and offers limited visibility into charging behaviour such as voltage, current, power, and energy usage. A promising alternative is to buffer locally generated and stored energy and deliver it to the charging point alongside the grid supply, so that charging power is no longer constrained by the grid connection alone.

This project presents the design and development of an IoT-enabled smart monitoring and energy-buffered charging system using embedded controllers and wireless communication modules, developed as a proof-of-concept for such an architecture. The system integrates an STM32F446RE microcontroller for real-time sensing and computation of charging parameters such as voltage, current, power, energy, and state of charge, and uses PWM-based control of a DC-DC converter to regulate charging. A supercapacitor is incorporated as an energy-buffering element to absorb short-duration fluctuations and transient surge currents, thereby improving charging stability and protecting the storage element. An ESP8266 module acts as an IoT gateway, transmitting processed data to a cloud-based dashboard for remote monitoring and analytics.

The prototype was evaluated under controlled low-voltage conditions and demonstrated stable real-time monitoring, embedded computation, PWM-based regulation, and supercapacitor-buffered operation. The work establishes the sensing, control, and monitoring foundation required for a scaled solar-assisted, battery-buffered EV charging point, which is identified as the target application for

future deployment.

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List of Abbreviations

Abbreviation	Description
ADC	Analog to Digital Converter
CC	Constant Current
CV	Constant Voltage
DC	Direct Current
DC-DC	Direct Current to Direct Current Converter
ESP8266	Wi-Fi Communication Module by Espressif
GND	Ground
I2C	Inter-Integrated Circuit Communication Protocol
IDE	Integrated Development Environment
INA219	Current and Voltage Sensor Module
IoT	Internet of Things
MCU	Microcontroller Unit
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
UART	Universal Asynchronous Receiver Transmitter
V / I / P / E	Voltage / Current / Power / Energy
Wi-Fi	Wireless Fidelity

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Chapter 1 – Introduction

1.1 Background

The growing adoption of battery-powered devices and electric mobility systems has increased the importance of efficient charging and monitoring mechanisms. Charging systems are commonly used in residential, industrial, and portable electronic applications. In many low-cost charging setups, the charging process operates without providing detailed information about voltage variation, current consumption, power utilization, or charging characteristics.

With the advancement of embedded systems and wireless communication technologies, it has become possible to develop low-cost intelligent monitoring solutions for charging applications. Real-time monitoring enables users to observe charging parameters continuously and analyse the behaviour of the system under different operating conditions.

The Internet of Things (IoT) has further enhanced the capability of embedded systems by enabling remote visualization and cloud-based monitoring of system parameters. IoT-enabled monitoring systems help in collecting, storing, and visualizing data in real time, thereby improving visibility into charging behaviour.

In addition to monitoring, energy-buffering techniques using supercapacitors have gained attention due to their high power density and fast charge-discharge capability. Supercapacitors can temporarily store energy and support short-duration fluctuations in the system. This makes them suitable for studying charging stability and transient energy behaviour.

The proposed project focuses on the design and implementation of an IoT-enabled smart monitoring and analysis system for charging applications. The system integrates sensing, embedded computation, PWM-based control, and wireless communication to observe charging behaviour in real time. A supercapacitor is included as an energy-buffering element to study its impact during charging operation.

1.2 Problem Statement

The accelerating adoption of electric vehicles (EVs) and rooftop solar installations has created a mismatch between the power an EV can accept during charging and the power a typical residential grid connection can supply. Conventional home charging draws entirely from the grid at a rate limited by the household service connection, which results in long charging durations and adds to peak demand on the distribution network. At the same time, solar-equipped households generate energy that is only partially self-consumed, with surplus commonly exported to the grid at low value.

A more effective approach is to buffer locally generated and stored energy and deliver it to the charging point in parallel with the grid supply, so that charging power is no longer constrained by the grid connection alone. In such an architecture, photovoltaic energy charges a stationary battery during the day, while a supercapacitor stage handles short-duration surge currents and transient fluctuations, protecting the battery and stabilising delivery to the charging point. This decoupling of charging power from the instantaneous grid limit can reduce EV charging time and relieve grid stress — an approach now validated in commercial battery-buffered charging systems [16][17][18].

Realising such a system requires reliable, real-time visibility into the underlying electrical parameters — voltage, current, power, energy, supercapacitor state, and battery state of charge — together with embedded control of the power-conversion stage. However, many low-cost charging setups operate without this instrumentation, offering limited insight into charging behaviour under fluctuating-source and energy-buffered conditions.

The present work addresses this gap by designing and implementing a proof-of-concept embedded and IoT-enabled platform that performs the sensing, computation, PWM-based converter control, supercapacitor buffering, and cloud-based monitoring required for such an architecture. The prototype is evaluated under controlled low-voltage conditions as a foundational building block, with the full solar-assisted, battery-buffered EV charging scenario identified as the target application for scaled deployment.

1.3 Objectives

The main objectives of the project are as follows:

- To design a smart charging monitoring system using embedded technology.
- To measure voltage and current parameters in real time.
- To compute power and energy during charging operation.
- To implement PWM-based control using the STM32 microcontroller.
- To transmit charging data to a cloud-based platform using the ESP8266 module.
- To study the effect of supercapacitor buffering on charging stability and transient behaviour.
- To visualize charging parameters through an IoT dashboard.

1.4 Scope of the Project

The scope of the project is limited to the development of a low-power prototype system for monitoring and analysing charging behaviour. The system includes sensing modules, embedded control, cloud communication, and a supercapacitor-based buffering arrangement.

The project does not involve industrial-scale charging infrastructure or high-voltage EV charging systems. The focus is primarily on embedded monitoring, computation, and analysis of charging parameters in a controlled prototype environment, serving as a building block toward the scaled solar-assisted, battery-buffered charging application described in the problem statement.

1.5 Organization of the Report

This report is organized into multiple chapters describing different aspects of the project work.

- Chapter 1 presents the introduction, objectives, and scope of the project.
- Chapter 2 discusses the literature survey related to charging systems, IoT monitoring, and supercapacitor applications.
- Chapter 3 explains the system architecture and hardware design.
- Chapter 4 describes the research methodology and computation techniques.
- Chapter 5 discusses implementation details and module integration.
- Chapter 6 presents the experimental observations and analysis.
- Chapter 7 concludes the work and discusses future enhancements.

Chapter 2 – Literature Review

2.1 Introduction

The rapid growth of battery-operated systems and electric mobility has increased research interest in charging technologies, energy storage systems, and intelligent monitoring solutions. Several studies have explored the use of embedded systems, IoT platforms, renewable energy integration, and supercapacitor-based energy storage mechanisms for improving charging and energy-management applications [1][3][4].

This chapter presents a review of existing work related to charging systems, supercapacitors, embedded monitoring systems, and IoT-enabled energy applications. The review also identifies the research gap that forms the basis for the proposed work.

2.2 Charging Systems and Energy Monitoring

Charging systems are widely used in applications such as portable electronics, battery-operated devices, and electric vehicles. Conventional charging systems are mainly designed to regulate the voltage and current supplied to the battery. However, many low-cost systems lack real-time monitoring and analytical capabilities [9].

Embedded monitoring systems have been increasingly adopted to provide visibility into charging parameters such as voltage, current, power, and energy consumption [8][9]. Real-time monitoring allows a better understanding of charging behaviour and system performance under varying operating conditions.

Several researchers have focused on integrating microcontrollers and wireless communication technologies into charging systems to enable remote observation and data visualization [13][14].

2.3 Supercapacitor-Based Energy Buffering

Supercapacitors are energy storage devices known for their high power density, fast charge-discharge capability, and long operational life. Unlike conventional batteries, supercapacitors store energy electrostatically rather than chemically [2].

Research studies have explored the use of supercapacitors as auxiliary energy storage devices in renewable energy systems, electric vehicles, and hybrid storage architectures [1][4]. Supercapacitors are often used to handle short-duration fluctuations, transient loads, and rapid energy-transfer applications.

Hybrid systems combining batteries and supercapacitors have demonstrated improved transient response and enhanced power-handling capability [3][5]. However, most existing research focuses on large-scale systems or advanced industrial applications.

The proposed work uses a supercapacitor primarily as an energy-buffering element to observe charging behaviour and short-term stability characteristics in a low-power prototype environment.

2.4 IoT-Based Monitoring Systems

The Internet of Things (IoT) has enabled the development of connected monitoring systems capable of collecting and transmitting real-time data over wireless networks [13][14]. IoT platforms are widely used in smart energy systems, industrial automation, and remote-monitoring applications.

ESP8266 and similar Wi-Fi modules are commonly used in embedded IoT systems due to their low cost and ease of integration [7]. Cloud platforms such as ThingSpeak and Blynk provide facilities for visualizing sensor data in graphical form [15].

Several studies have demonstrated the use of IoT for monitoring environmental parameters, energy consumption, and device status [13]. However, many systems focus only on data visualization without integrating embedded computation and charging analysis.

In the proposed system, IoT is used for remote monitoring and visualization of charging parameters rather than direct control of the charging process.

2.5 Embedded Control and PWM Techniques

Embedded controllers are widely used in power electronics and charging applications for sensing, computation, and control [9][12]. Microcontrollers such as the STM32 provide high-speed processing capability and support the interfaces required for sensor integration and communication [6][12].

PWM (Pulse Width Modulation) is commonly used in DC-DC converters to regulate the voltage and current supplied to the load [9][11]. By varying the duty cycle of the PWM signal, the effective output power can be controlled efficiently.

Several charging systems implement Constant Current (CC) and Constant Voltage (CV) charging techniques using PWM-based control methods [10][11]. Embedded control also enables the implementation of protection mechanisms and monitoring functions.

The proposed work utilizes the STM32 for sensor acquisition, embedded computation, and PWM-based regulation of the charging system.

2.6 Research Gap

From the literature survey, it is observed that many existing works focus either on advanced charging infrastructure, renewable energy integration, or industrial-scale energy-management systems. Limited work is available on simple and modular prototype systems that combine embedded sensing and computation, PWM-based charging control, IoT-enabled monitoring, and supercapacitor-based buffering observation in a single low-cost implementation.

The proposed project attempts to address this gap by developing a practical prototype for monitoring and analysing charging behaviour using embedded systems and IoT technologies.

2.7 Summary

This chapter reviewed existing research related to charging systems, supercapacitors, IoT monitoring, and embedded control techniques. The review highlights the importance of real-time monitoring and embedded computation in charging applications. Based on the identified research gap, the proposed system focuses on developing a smart monitoring and analysis platform using STM32 and ESP8266 modules along with a supercapacitor-based buffering arrangement.

Chapter 3 – System Design and Architecture

3.1 Introduction

The proposed system is designed to monitor and analyse charging behaviour using embedded sensing, computation, and wireless communication techniques. The architecture combines hardware modules responsible for sensing, control, power regulation, communication, and energy buffering.

The system is implemented as a low-power prototype to study charging characteristics in a controlled environment. The overall design focuses on modularity, simplicity, and real-time monitoring capability.

3.2 System Overview

The system consists of the following major modules: Power Supply Unit; DC-DC Converter; Supercapacitor Buffer; Battery / Load Module; Voltage and Current Sensing Unit; STM32 Embedded Controller; ESP8266 IoT Communication Module; and Cloud Monitoring Dashboard.

The interaction between these modules enables the acquisition, processing, and transmission of charging data for real-time monitoring and analysis.

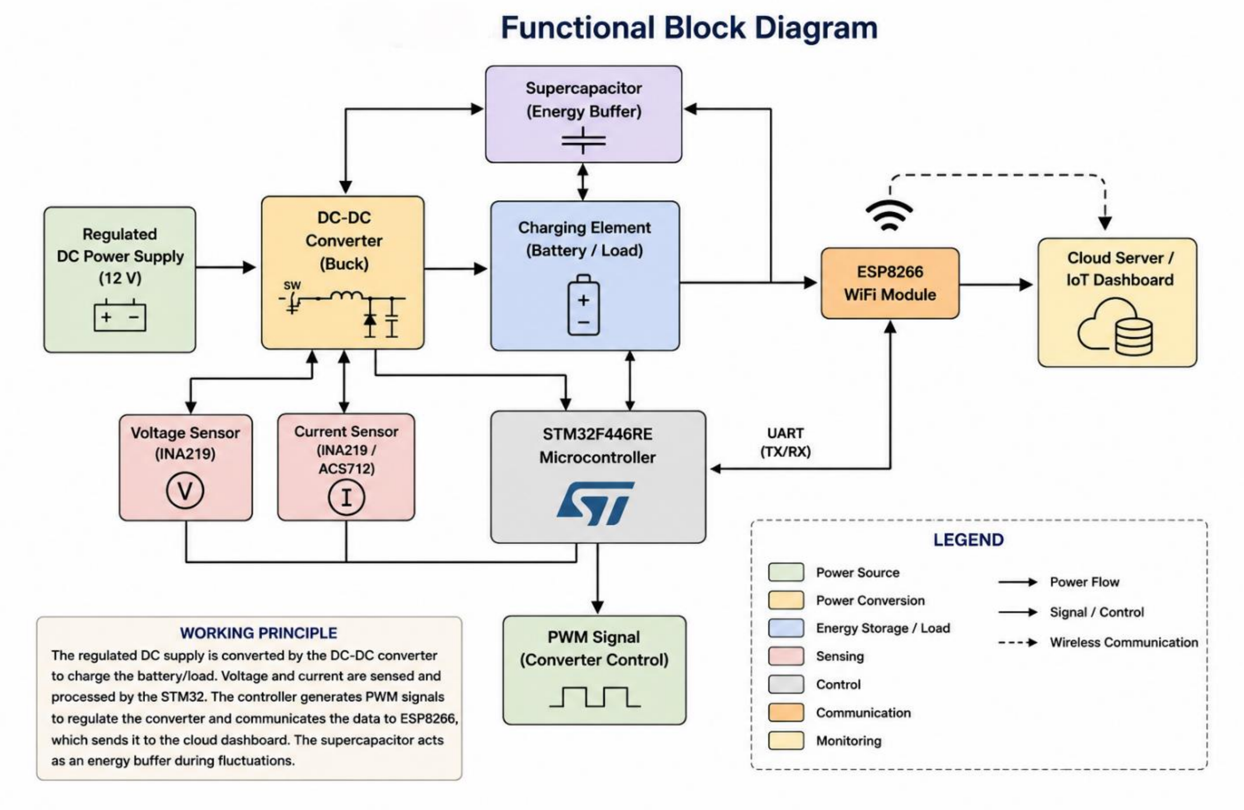

Figure 3.1 Functional Block Diagram

3.3 System Architecture

The proposed system architecture is designed to integrate sensing, embedded computation, power regulation, and IoT-based monitoring into a single modular platform. The system consists of a regulated power source, DC-DC converter, supercapacitor buffer, sensing unit, STM32F446RE controller, ESP8266 communication module, and a cloud-based monitoring dashboard.

Voltage and current sensors continuously acquire charging parameters and transmit the measured values to the STM32 controller. The controller performs real-time computations such as power, energy, and charging analysis, while also generating PWM signals for converter regulation. The processed data is transmitted to the ESP8266 module through UART communication and uploaded to the cloud dashboard for remote visualization and monitoring.

The input power source provides regulated DC power to the charging system. The DC-DC converter regulates the voltage supplied to the battery or load. A supercapacitor is incorporated as an intermediate energy-buffering element to observe charging behaviour under varying conditions. Voltage and current sensors continuously measure electrical parameters from the system. The STM32 controller processes sensor data, computes charging parameters, and controls the converter through PWM signals. The processed data is transmitted to the ESP8266 module for cloud-based monitoring.

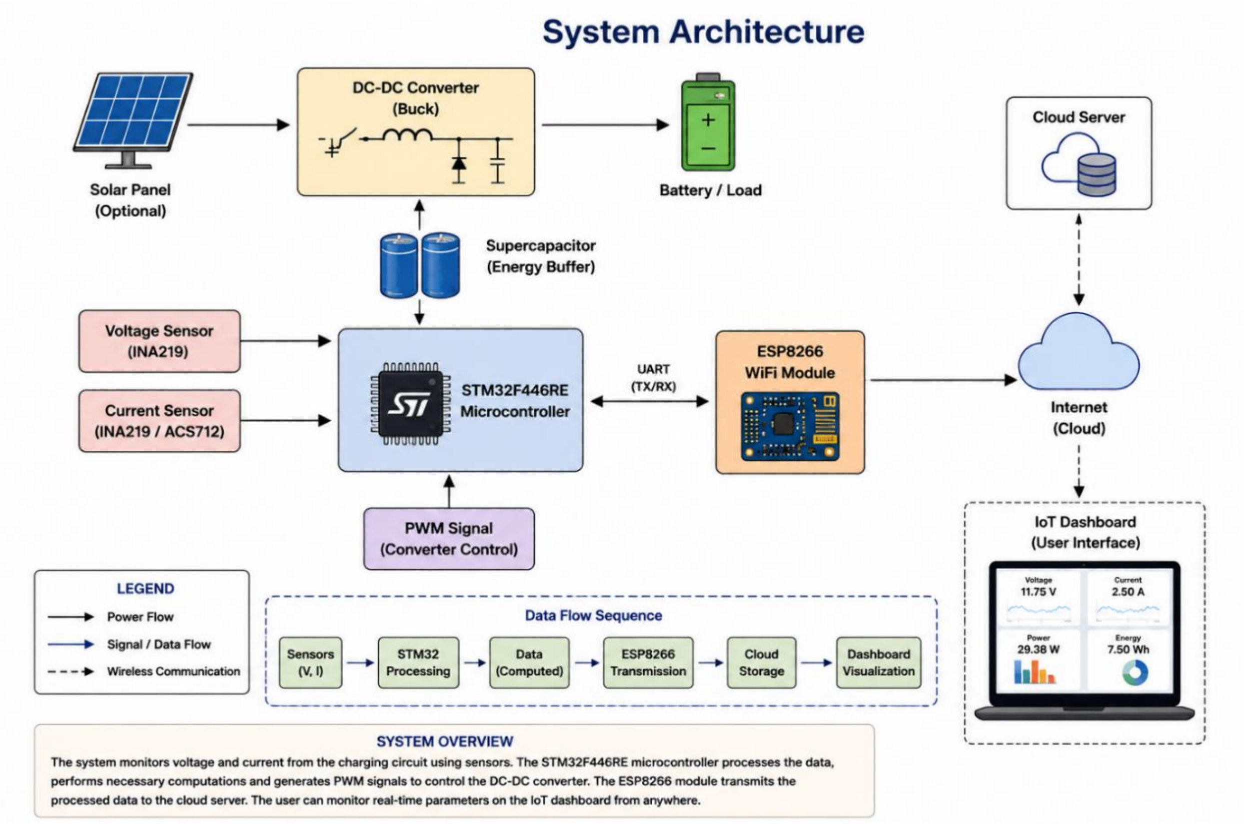

Figure 3.2 Overall System Architecture

3.4 Hardware Modules

3.4.1 STM32F446RE Microcontroller

The STM32F446RE microcontroller acts as the central processing and control unit of the system. It performs sensor data acquisition, power and energy computation, PWM signal generation, and communication with the ESP8266. The controller is selected due to its high processing capability, built-in peripherals, and suitability for embedded control applications.

3.4.2 ESP8266 Communication Module

The ESP8266 module provides wireless communication functionality. It receives processed data from the STM32 controller through UART communication and transmits the data to a cloud-based dashboard using Wi-Fi connectivity. The module enables remote visualization of charging parameters such as voltage, current, and power.

3.4.3 Voltage and Current Sensors

Voltage and current measurements are performed using sensor modules connected to the STM32 controller. The sensing unit performs measurement of charging voltage, measurement of charging current, and data transmission to the controller. These measurements are used for computation and monitoring purposes.

3.4.4 DC-DC Converter

The DC-DC converter regulates the input power supplied to the battery or load. The converter operates based on PWM control signals generated by the STM32 controller, enabling controlled charging operation by adjusting voltage and current levels.

3.4.5 Supercapacitor Buffer

A supercapacitor is used as a temporary energy storage element within the charging system. Due to its fast charge-and-discharge capability, the supercapacitor helps in handling short-duration fluctuations and transient variations. The supercapacitor is incorporated primarily for observation and analysis of buffering behaviour during charging.

3.4.6 Battery / Load Module

The battery or equivalent load acts as the charging element in the prototype system. The charging behaviour of the battery is monitored using embedded sensing and cloud visualization.

3.4.7 Power Supply Unit

A regulated DC power source is used to supply energy to the prototype system. The supply voltage is maintained above the converter output voltage and within safe operating limits to ensure stable buck-mode operation of all hardware modules.

3.4.8 Hardware Setup Implementation

The hardware setup diagram illustrates the physical arrangement and interconnection of the major components used in the proposed smart charging monitoring system. The setup includes the STM32F446RE controller, ESP8266 communication module, voltage and current sensing units, DC-DC converter, supercapacitor buffer, and regulated power supply.

The STM32 controller performs real-time sensing, computation, and PWM-based control, while the ESP8266 module enables wireless communication with the cloud dashboard. Proper grounding, stable power connections, and reliable sensor interfacing are maintained to ensure accurate measurement and stable operation of the system. The hardware arrangement is designed using a modular approach, allowing easy testing, debugging, and future expansion of the prototype system.

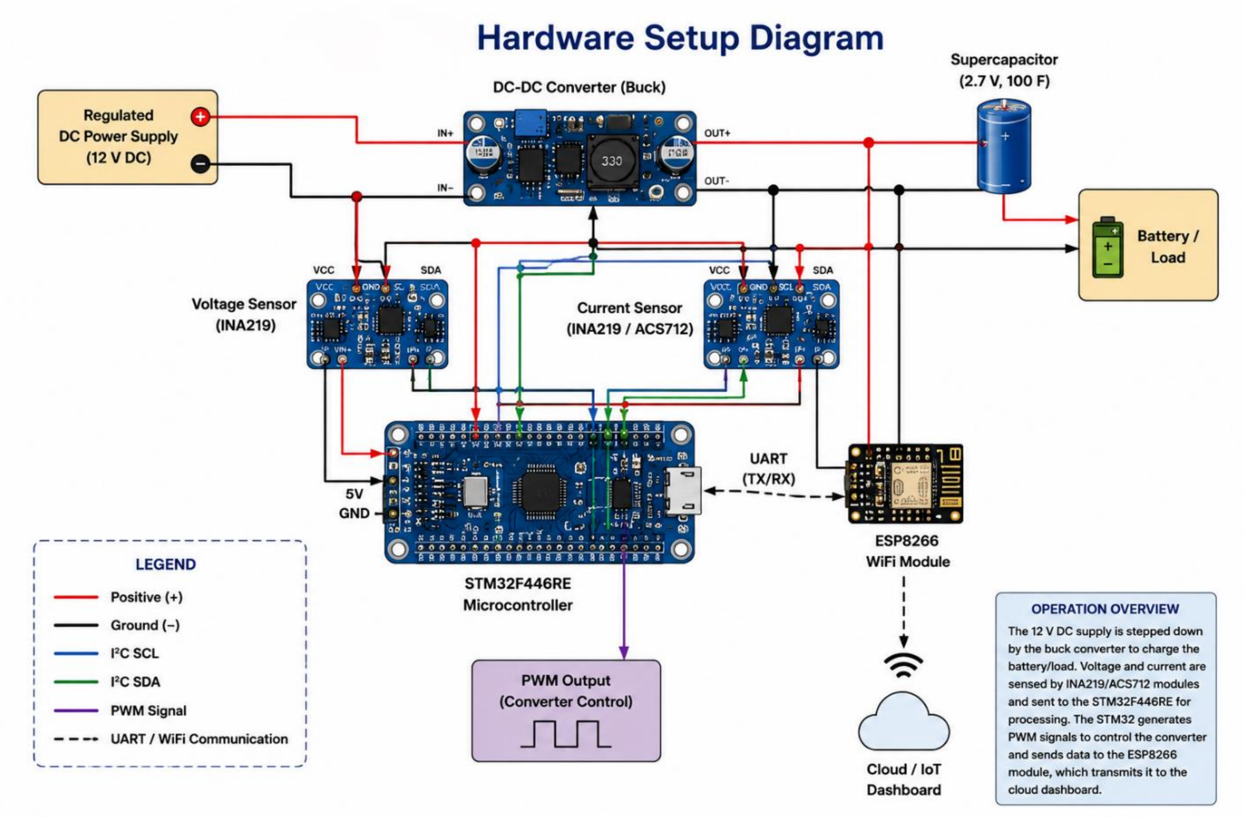

Figure 3.3 Hardware Set-up Diagram of Proposed System

3.4.9 Hardware Circuit Diagram

The hardware circuit diagram represents the electrical interconnection between the major components of the proposed charging monitoring system. It includes the regulated power supply, DC-DC converter, supercapacitor buffer, sensing modules, STM32F446RE controller, and ESP8266 communication module.

The circuit is designed to enable real-time sensing, embedded computation, PWM-based converter control, and wireless transmission of charging parameters. Proper grounding and signal interfacing are maintained to ensure stable operation and accurate measurement of voltage and current values during system operation.

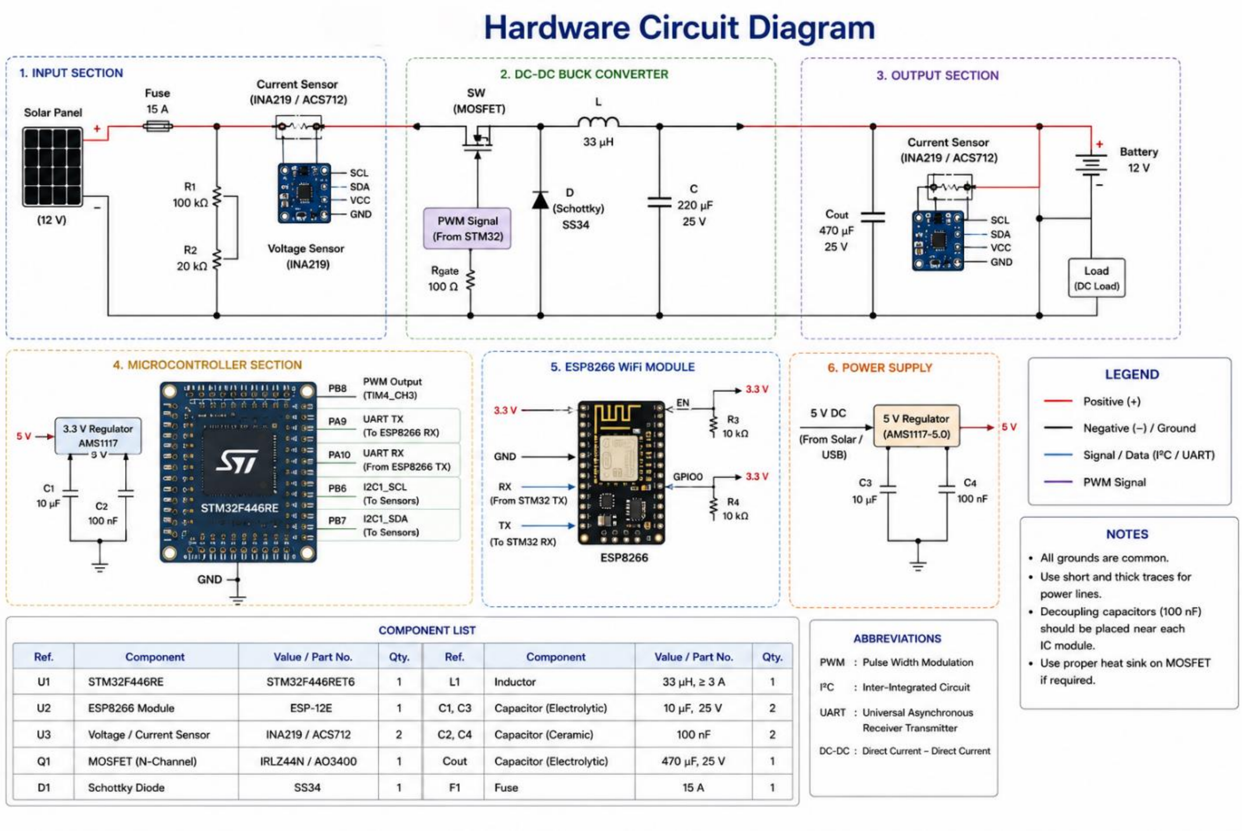

Figure 3.4 Hardware Circuit Diagram of Proposed System

3.5 Communication Interfaces

3.5.1 UART Communication

UART communication is used between the STM32 and ESP8266 modules for serial data transmission. Sensor data and computed values are transmitted in serial format.

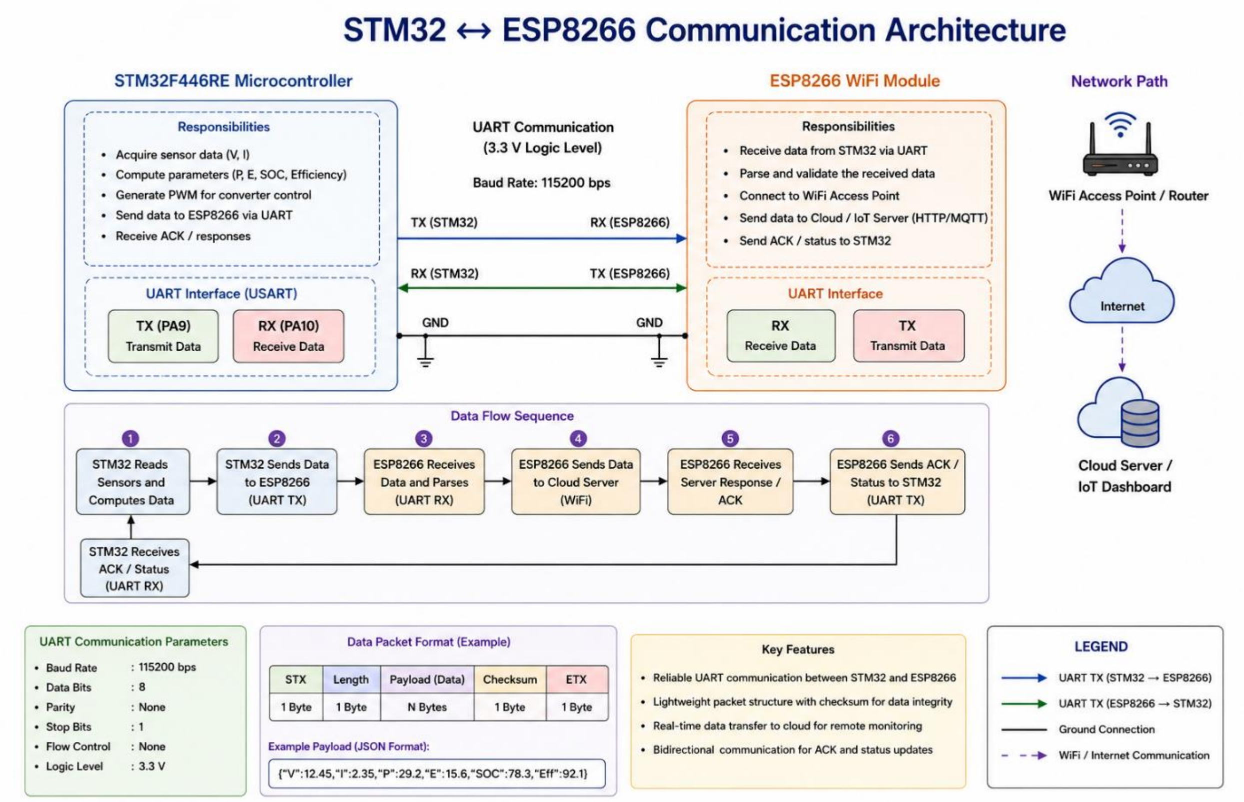

Figure 3.5 STM32 and ESP8266 Communication Architecture

3.5.2 PWM Interface

PWM signals generated by the STM32 are used to control the operation of the DC-DC converter. The duty cycle of the PWM signal determines the effective output voltage and current.

3.5.3 Sensor Interface

Voltage and current sensors are connected to the STM32 through analog or digital interfaces. The controller continuously reads sensor values during operation.

3.5.4 Wi-Fi Communication

The ESP8266 module connects to a wireless network and transmits charging data to the cloud dashboard for visualization and monitoring.

3.6 System Operation

The system operates in the following sequence:

1. The regulated DC supply provides power to the charging system.
2. The DC-DC converter regulates the voltage supplied to the load.
3. Voltage and current sensors measure charging parameters.
4. The STM32 acquires sensor data and performs computations.
5. PWM signals are generated for converter control.
6. Computed data is transmitted to the ESP8266 through UART communication.

7. The ESP8266 uploads the data to the cloud dashboard.
8. Charging behaviour is monitored and analysed in real time.

3.7 Design Considerations

The system is designed based on the following considerations: low-cost implementation; modular hardware structure; real-time monitoring capability; safe operating voltage levels; ease of integration and testing; wireless communication support; and a simple, scalable architecture.

3.8 Summary

This chapter described the system architecture, hardware modules, interfaces, and operational flow of the proposed charging monitoring system. The architecture combines embedded control, sensing, communication, and energy-buffering techniques to enable real-time monitoring and analysis of charging behaviour.

Chapter 4 – Research Methodology

4.1 Introduction

This chapter describes the methodology adopted for the development of the proposed smart charging monitoring system. The methodology includes system design, hardware integration, embedded programming, data acquisition, computation, communication, and monitoring. The overall approach focuses on implementing a modular prototype capable of measuring and analysing charging parameters in real time using embedded systems and IoT technologies.

4.2 Methodology Overview

The development methodology is divided into the following stages: Problem Identification; System Design; Hardware Selection; Sensor Integration; Embedded Programming; Communication Setup; Cloud Monitoring; and Testing and Analysis. Each stage contributes to the implementation of the complete monitoring system.

4.3 Problem Identification

Many low-cost charging systems provide limited visibility into charging behaviour and electrical parameters. Users are often unable to monitor voltage, current, power consumption, and charging variations in real time. The project aims to address this limitation by integrating sensing, embedded computation, and IoT-based monitoring into a single prototype platform.

4.4 System Design Methodology

The system architecture is designed using a modular approach to simplify integration and testing. The design methodology includes selection of a suitable embedded controller, identification of sensing requirements, integration of communication modules, implementation of power regulation, and inclusion of a supercapacitor buffering arrangement. The modular design allows independent testing of individual modules before complete integration.

4.5 Hardware Selection

The hardware components are selected based on simplicity, cost-effectiveness, and compatibility with embedded applications.

4.5.1 STM32F446RE Controller

The STM32F446RE microcontroller is selected as the main processing unit due to its high processing speed, PWM generation capability, multiple communication interfaces, and ADC support for sensor acquisition.

4.5.2 ESP8266 Module

The ESP8266 is selected for wireless communication because of its built-in Wi-Fi support, ease of integration, low cost, and compatibility with cloud platforms.

4.5.3 Sensors

Voltage and current sensors are selected to continuously monitor charging parameters. The sensing unit enables voltage measurement, current measurement, and data acquisition for computation.

4.5.4 Supercapacitor

A supercapacitor is included as an energy-buffering element to study its behaviour during charging operation. Its high power density and fast charge-discharge capability make it suitable for observing short-duration fluctuations.

4.6 Data Acquisition Methodology

Voltage and current data are acquired continuously using sensors connected to the STM32 controller. The acquired data is sampled periodically, converted into digital values, and processed for further computation. The sensor data forms the basis for power and energy calculations.

4.7 Embedded Computation

The STM32 controller performs real-time computations using sensor data acquired from the charging system. These computations are used for monitoring, analysis, and PWM-based control.

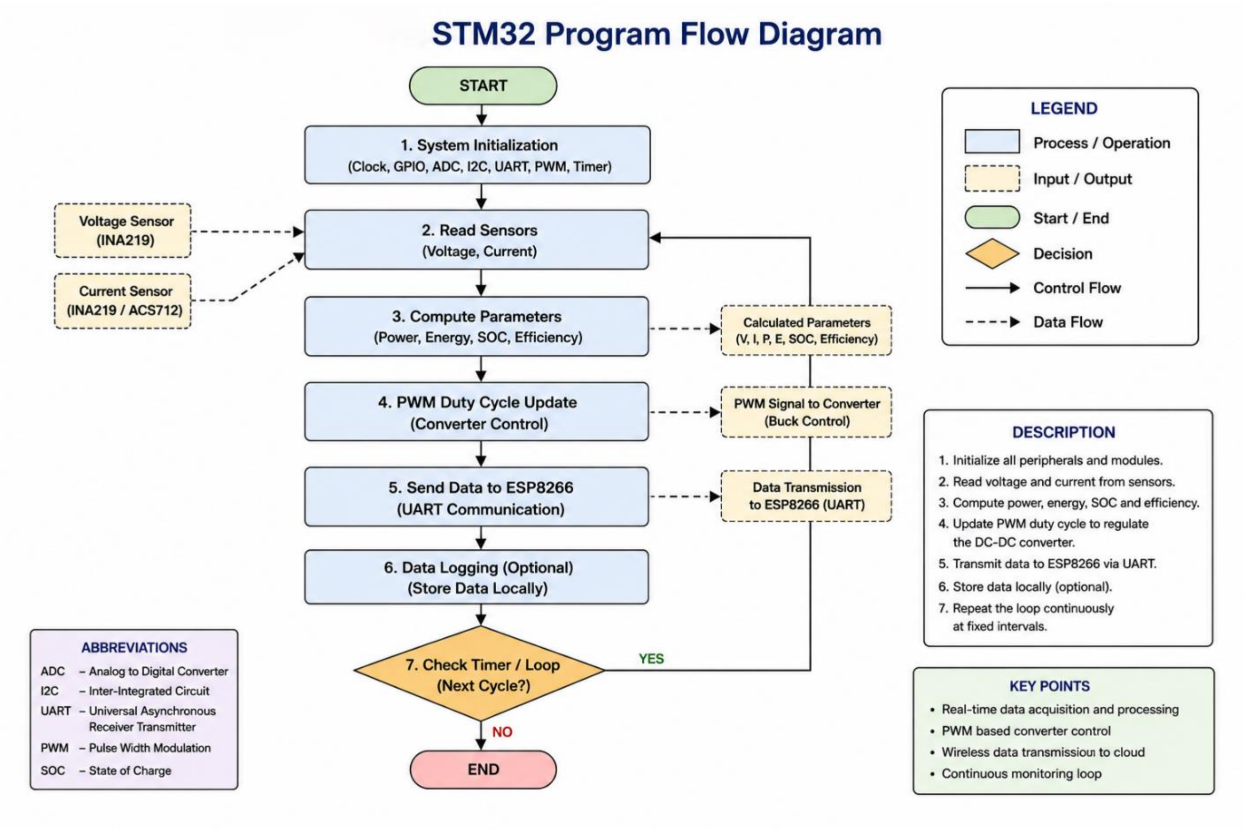

Figure 4.1 STM32 Program Flow for Charging Computation

4.7.1 Power Calculation

The instantaneous power delivered to the charging system is calculated using:

$$P = V \times I$$

where P is power (watts), V is voltage (volts), and I is current (amperes). The STM32 controller continuously acquires voltage and current data through sensors and computes power in real time.

4.7.2 Energy Calculation

The total energy transferred during charging is estimated by integrating power over time, $E = \int P\,dt$. For discrete embedded implementation:

$$E = P \times t$$

where E is energy, P is power, and t is the time duration. The cumulative energy value is used for charging analysis and system monitoring.

4.7.3 Supercapacitor Energy Estimation

The energy stored in the supercapacitor is estimated using:

$$E = \tfrac{1}{2} C V^2$$

where C is capacitance (farads) and V is the voltage across the supercapacitor. This computation helps in estimating the buffering capability of the supercapacitor during transient conditions.

4.7.4 PWM Duty Cycle Calculation

The PWM duty cycle is calculated using:

$$D = (V_{out} / V_{in}) \times 100$$

where D is the duty cycle (%), V_{out} is the required output voltage, and V_{in} is the input voltage. The STM32 dynamically adjusts the PWM duty cycle to regulate converter operation.

4.7.5 State of Charge Estimation

The battery State of Charge (SOC) is estimated using:

$$SOC = (Q_{charged} / Q_{max}) \times 100$$

where $Q_{charged}$ is the current charge level and Q_{max} is the maximum battery capacity. SOC estimation helps in monitoring charging progress and charging-state transitions.

4.8 PWM-Based Control Methodology

PWM (Pulse Width Modulation) is used to regulate the operation of the DC-DC converter. The STM32 controller generates PWM signals with varying duty cycles to regulate output voltage, control charging current, and maintain stable operation. Duty-cycle adjustment enables efficient control of the power delivered to the load.

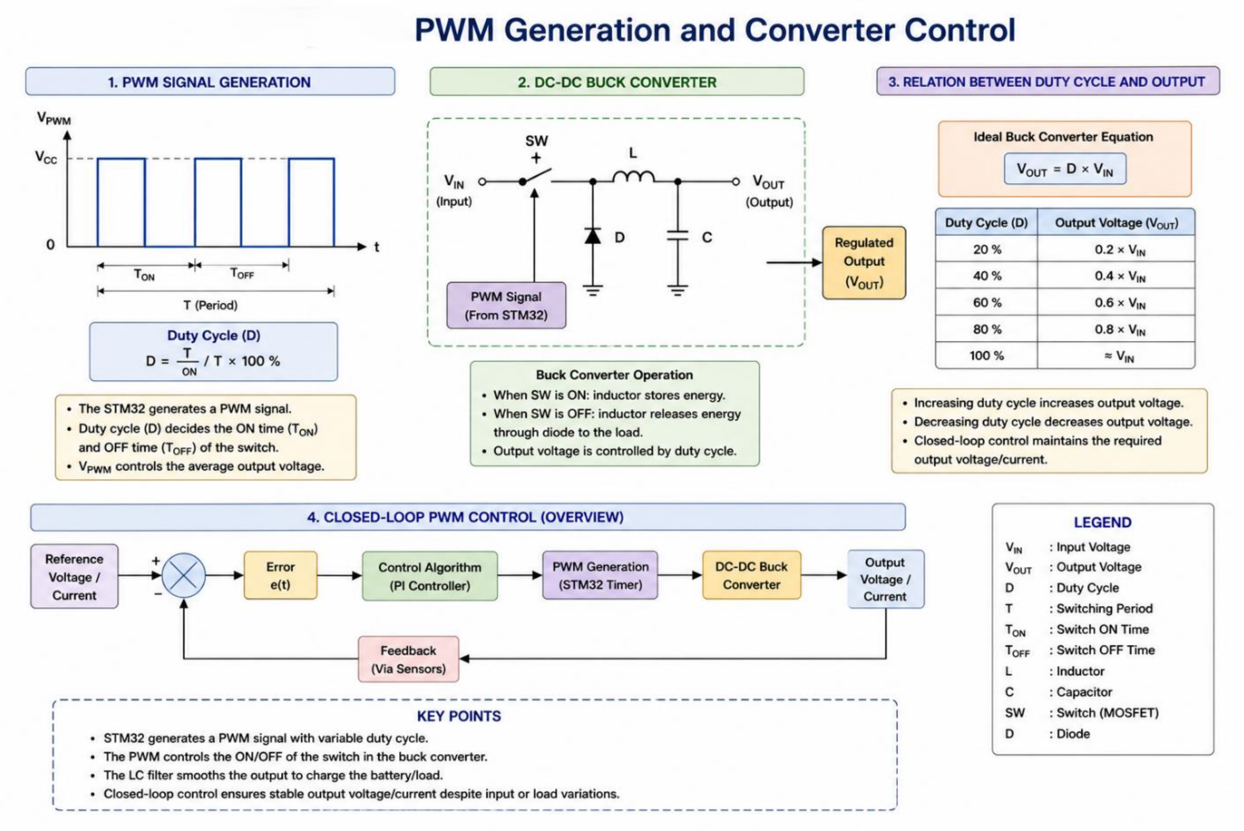

Figure 4.2 PWM Generation and DC-DC Converter Control

4.9 Communication Methodology

Communication between the STM32 and ESP8266 is established using UART serial communication. The communication process includes sensor data acquisition, data formatting by the STM32, serial transmission to the ESP8266, and Wi-Fi transmission to the cloud platform. This enables remote monitoring and visualization of charging parameters.

4.10 Cloud Monitoring Methodology

The cloud platform is used for visualizing charging data in graphical form.

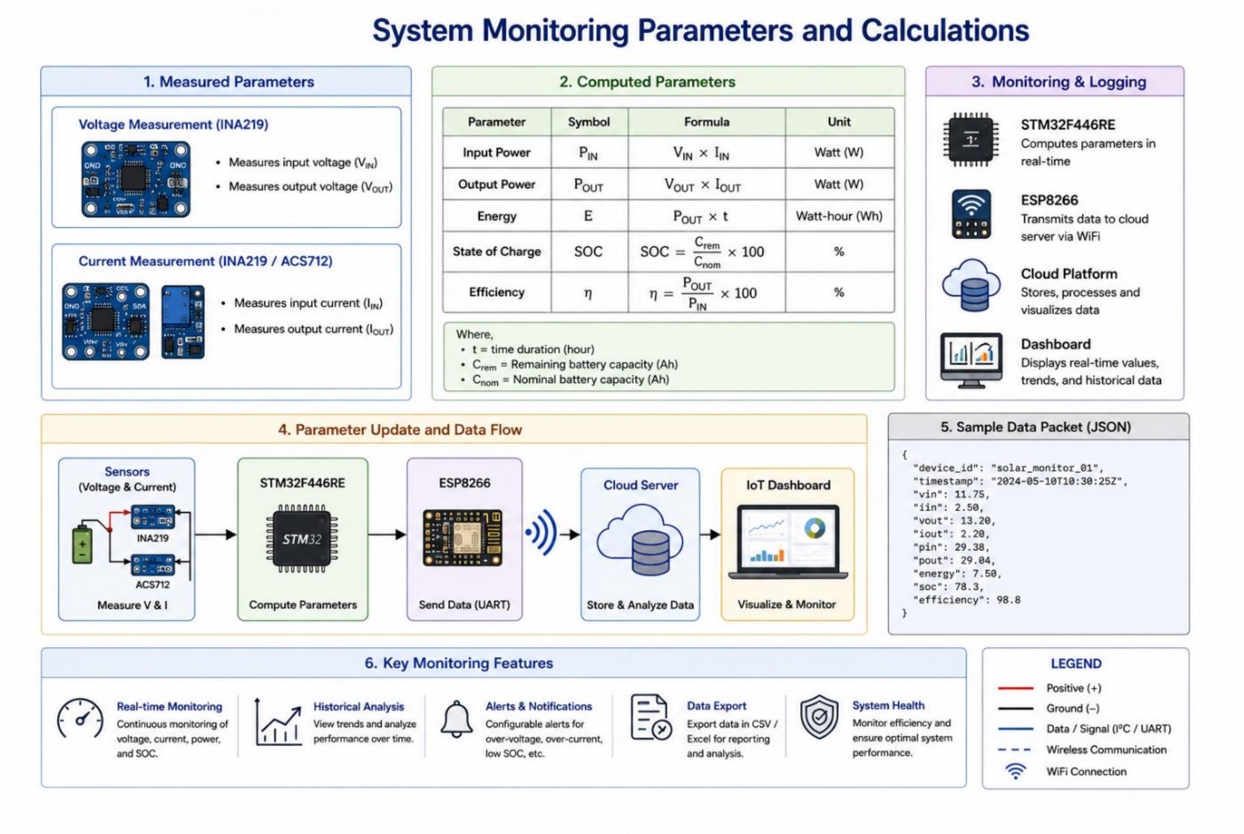

Figure 4.3 Monitoring Parameters and Embedded Calculations

The monitoring dashboard displays voltage variation, current variation, power values, and charging trends over time. Cloud-based monitoring enables observation and analysis of system behaviour remotely.

4.11 Experimental Methodology

The experimental setup is developed to observe charging behaviour under controlled conditions. The testing procedure includes operation of the charging system under a regulated DC supply, monitoring of voltage and current values, observation of supercapacitor behaviour, and visualization of charging parameters through the dashboard. Comparative observations may also be performed with and without the supercapacitor buffering arrangement.

4.12 Firmware Execution Flow

The firmware execution flow diagram illustrates the sequence of operations performed by the STM32F446RE microcontroller during system operation. The firmware begins with system initialization, including configuration of GPIOs, ADC channels, timers, UART communication, and PWM modules.

After initialization, the controller continuously executes a loop in which voltage and current sensor data are acquired and processed. The firmware computes parameters such as power, energy, and charging state, and generates PWM signals to regulate the DC-DC converter. The processed data is then transmitted to the ESP8266 communication module for cloud-based monitoring. The firmware also includes basic control and protection logic for handling abnormal operating conditions, ensuring stable and reliable operation of the charging system.

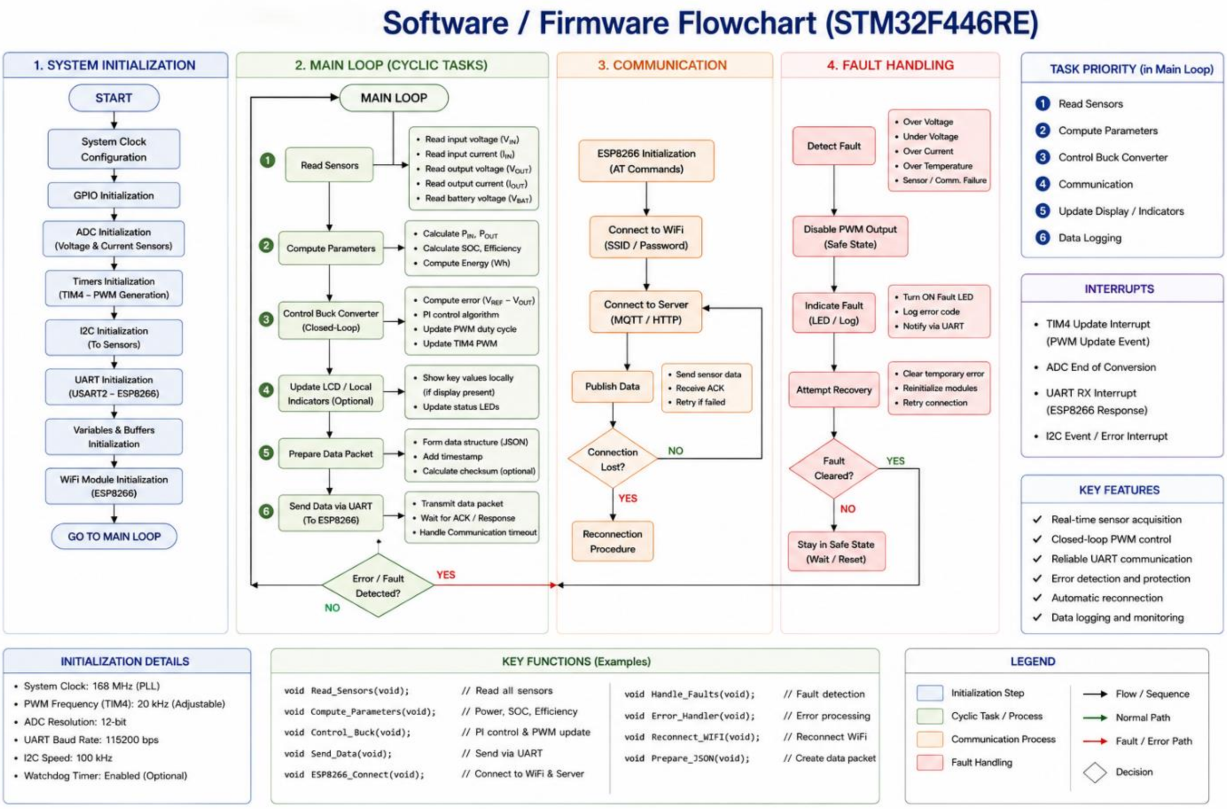

Figure 4.4 Software and Firmware Flowchart

4.13 Summary

This chapter presented the methodology adopted for the implementation of the proposed smart charging monitoring system. The methodology includes system design, sensing, embedded computation, PWM control, communication, and cloud-based monitoring. The next chapter discusses the implementation details and integration of the hardware and software modules used in the project.

Chapter 5 – Implementation

5.1 Introduction

This chapter describes the practical implementation of the proposed IoT-enabled smart charging monitoring system. The implementation includes hardware integration, embedded firmware development, sensing mechanisms, PWM-based control, wireless communication, and cloud-based monitoring. The system is developed as a low-power prototype to study charging behaviour using embedded systems and IoT technologies.

5.2 Hardware Implementation

The hardware implementation consists of sensing, control, communication, and energy-buffering modules integrated into a single prototype setup. The major hardware components used are the STM32F446RE development board, ESP8266 Wi-Fi module, voltage and current sensors, DC-DC converter, supercapacitor, battery / load module, regulated DC power supply, and connecting wires and PCB interfaces. The hardware modules are interconnected to enable real-time acquisition and processing of charging parameters.

5.3 STM32 Firmware Development

The STM32F446RE microcontroller acts as the central controller of the system. Firmware development is carried out using embedded C programming. The STM32 continuously reads voltage and current sensor values and processes them for charging analysis. The firmware performs sensor data acquisition, power and energy computation, PWM signal generation, serial communication with the ESP8266, and monitoring and control operations.

5.4 Sensor Interfacing

Voltage and current sensors are interfaced with the STM32 controller for real-time monitoring of charging parameters. The sensor interfacing process includes analog signal acquisition, ADC conversion, data calibration, and parameter computation. The acquired sensor values are used for calculating power, energy, charging efficiency, and State of Charge (SOC). The STM32 periodically samples sensor data during charging operation.

5.5 PWM-Based Converter Control

PWM control is implemented using the timer peripherals available in the STM32. The PWM signal controls the switching operation of the DC-DC converter. By varying the duty cycle of the PWM signal, the effective output voltage and charging current can be regulated. The PWM duty cycle is calculated using $D = (V_{out} / V_{in}) \times 100$. The STM32 dynamically adjusts the duty cycle based on sensor measurements and operating conditions.

5.6 Supercapacitor Integration

A supercapacitor is integrated into the charging system as an energy-buffering element. The supercapacitor performs temporary energy storage, handling of short-duration fluctuations, support during transient conditions, and buffering of voltage variations. The stored energy in the supercapacitor is estimated using $E = \frac{1}{2} C V^2$. The STM32 monitors the supercapacitor voltage continuously during operation.

5.7 Communication Implementation

Communication between the STM32 and ESP8266 is implemented using UART serial communication. The ESP8266 module connects to the Wi-Fi network and uploads charging data to the cloud dashboard. The communication process includes acquisition of sensor values, formatting of computed data, UART transmission to the ESP8266, and wireless transmission to the cloud platform.

5.8 IoT Dashboard Implementation

The IoT dashboard is used for visualization and monitoring of charging parameters in real time.

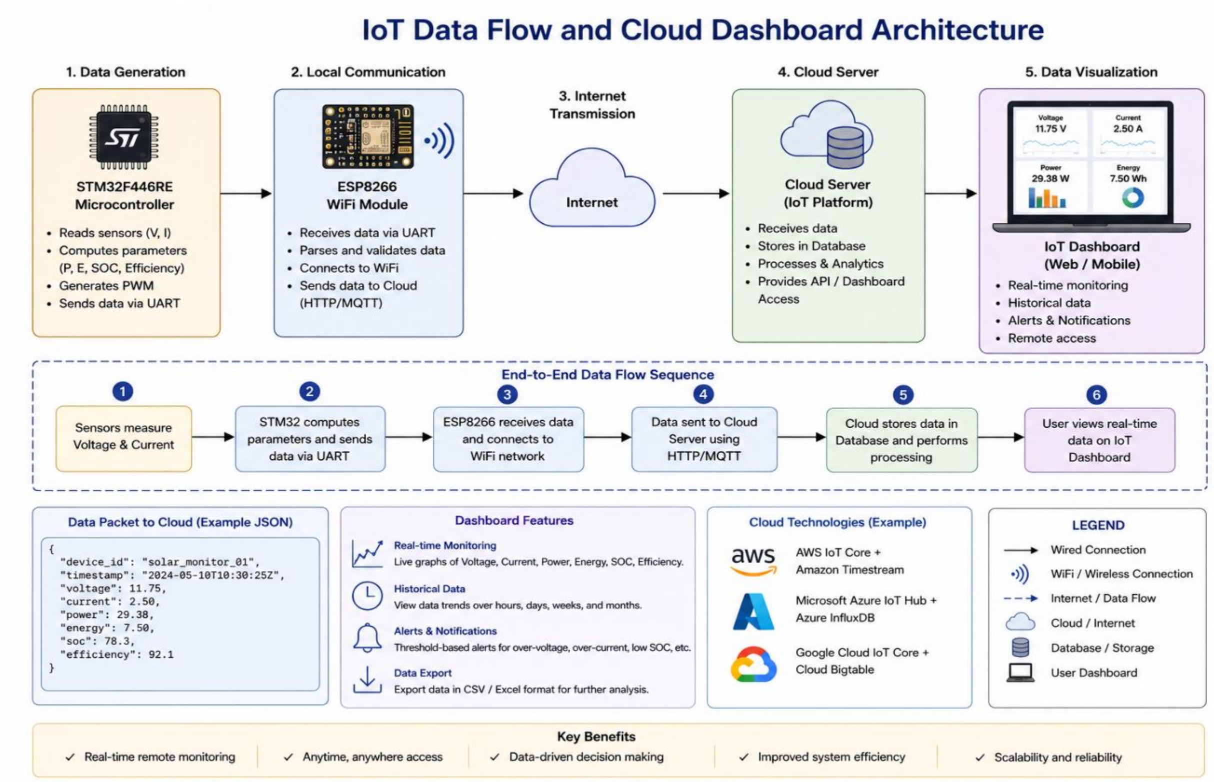

Figure 5.1 IoT Data Flow and Cloud Dashboard Architecture

The dashboard displays voltage values, current values, power variation, energy accumulation, and charging trends. The cloud platform enables remote observation of system behaviour during charging operation.

5.9 Experimental Setup

The prototype system is implemented under controlled low-voltage operating conditions.

Figure 5.2 Real-Time IoT Monitoring Dashboard

The experimental setup includes a regulated DC power source, charging converter, sensor modules, embedded controller, communication module, supercapacitor buffer, and monitoring dashboard. The setup is used to observe charging behaviour and parameter variations under different operating conditions.

5.10 System Operation Sequence

The implemented system operates in the following sequence: the regulated power supply energizes the charging system; the DC-DC converter regulates charging voltage; voltage and current sensors continuously acquire charging parameters; the STM32 processes sensor data and performs computations; PWM signals are generated to control converter operation; processed data is transmitted to the ESP8266 using UART communication; the ESP8266 uploads data to the cloud dashboard; and charging parameters are visualized and analysed in real time.

5.11 Implementation Challenges

During implementation, the following practical challenges were observed: sensor calibration accuracy; noise in analog measurements; stable PWM generation; communication synchronization; voltage-fluctuation handling; and hardware-integration complexity. These challenges were addressed through modular testing and controlled experimentation.

5.12 Summary

This chapter discussed the practical implementation of the proposed smart charging monitoring system. The implementation integrates sensing, embedded computation, PWM control, wireless communication, and cloud-based monitoring into a single prototype platform. The next chapter presents the experimental observations, computations, and performance analysis of the implemented system.

Chapter 6 – Results and Analysis

6.1 Introduction

This chapter presents the experimental observations and analysis of the proposed IoT-enabled smart charging monitoring system. The analysis includes monitoring of voltage, current, power, energy utilization, PWM control behaviour, and supercapacitor buffering characteristics. The experimental setup was operated under controlled low-voltage conditions to study charging behaviour and observe parameter variations in real time.

6.2 Experimental Setup

The experimental setup consists of the STM32F446RE controller, ESP8266 Wi-Fi module, voltage and current sensors, DC-DC converter, supercapacitor buffer, battery / load module, and regulated DC power supply. The STM32 controller continuously acquires sensor data and performs computations related to charging behaviour. The processed values are transmitted to the cloud dashboard for monitoring and visualization.

6.3 Voltage Monitoring Analysis

Voltage monitoring was performed continuously during charging operation. The observed voltage values were used for converter regulation, PWM duty-cycle adjustment, charging-state monitoring, and supercapacitor observation. During normal operation, the charging voltage remained within the expected operating range, with temporary variations observed during simulated fluctuation conditions. To permit correct buck-mode operation, the regulated input bus is maintained above the converter output (charging) voltage.

Table 6.1 Voltage Observation

Observation Parameter	**Typical Value**
Input Supply (DC Bus) Voltage	18 V
Converter Output (Charging) Voltage	14.4 V
Battery Voltage Range	11.8 V – 14.4 V
Supercapacitor Voltage	2.0 V – 2.7 V

The STM32 continuously monitored voltage values to ensure stable system operation.

6.4 Current Monitoring Analysis

The charging current was measured continuously using the current-sensing module.

Table 6.2 Current Observation

Observation Parameter	Typical Value
Initial Charging Current	2.5 A
Peak Current Observed	3 A
Average Charging Current	2.5 A

The charging current remained relatively stable during normal operation. Variations in current were observed during transient operating conditions and simulated supply fluctuations.

6.5 Power Computation Analysis

The STM32 controller computed the instantaneous charging power using P = V × I. For a charging voltage of 14.4 V and a current of 2.5 A:

$$P = 14.4 \times 2.5 = 36\ W$$

This indicates that the charging system delivers up to 36 W of power during operation.

Table 6.3 Power Observation

Voltage	Current	Computed Power
14.4 V	2.5 A	36 W
14.4 V	2.0 A	28.8 W
13.2 V	1.5 A	19.8 W

The computed power values were transmitted to the IoT dashboard for visualization.

6.6 Energy Analysis

The STM32 estimated cumulative charging energy using E = P × t. For an average power of 22.5 W and a charging duration of 20 minutes:

$$E = 22.5 \times (20 / 60) = 7.5\ Wh$$

The cumulative energy increased continuously during charging operation.

Table 6.4 Energy Observation

Duration	Average Power	Energy Delivered
5 min	22.5 W	1.875 Wh
10 min	22.5 W	3.75 Wh
20 min	22.5 W	7.5 Wh

6.7 PWM Duty Cycle Analysis

The PWM duty cycle was dynamically adjusted by the STM32 based on operating conditions, using $D = (V_{out} / V_{in}) \times 100$. For an input voltage of 18 V and a required output voltage of 14.4 V:

$$D = (14.4 / 18) \times 100 = 80\%$$

Table 6.5 PWM Observation

Input Voltage	Output Voltage	Duty Cycle
18 V	14.4 V	80%
16 V	14.4 V	90%
14 V	14.4 V	>100% (not possible)

When the duty-cycle requirement exceeded practical limits (input voltage approaching the required output), the supercapacitor buffer supported the charging operation during the transient.

6.8 Supercapacitor Analysis

The supercapacitor was used as a temporary energy-buffering element during voltage-fluctuation conditions. The stored energy was estimated using $E = ½\ C\ V^2$. For a capacitance of 100 F at 2.7 V:

$$E = ½ \times 100 \times 2.7^2 = 364.5\ J$$

The supercapacitor provided temporary buffering support during simulated transient conditions. It should be noted that 364.5 J corresponds to roughly 0.1 Wh; the supercapacitor therefore supports short-duration surge currents and transients rather than bulk energy delivery, while the battery provides sustained energy.

Table 6.6 Supercapacitor Observation

Capacitance	Voltage	Stored Energy
100 F	2.7 V	364.5 J
100 F	2.0 V	200 J

6.9 Charging Stability and Efficiency Analysis

Charging efficiency was estimated using $\eta = (E_{out} / E_{in}) \times 100$. For an input energy of 100 Wh and an output energy of 92 Wh:

$$\eta = (92 / 100) \times 100 = 92\%$$

A comparative observation was made with and without the supercapacitor buffer. The buffered configuration showed reduced voltage and current fluctuation and more stable regulation during transient conditions, which improved the consistency of energy delivery to the load.

Table 6.7 Charging Stability Observation

Operating Condition	Effective Efficiency
Without buffering (with transient fluctuation)	89%
With supercapacitor buffering	92%

The observed improvement is attributed to reduced fluctuation and better converter regulation rather than to any increase in the intrinsic conversion efficiency of the converter; a supercapacitor smooths transients but does not by itself raise steady-state conversion efficiency. In a fully buffered architecture, routing energy through an additional storage stage also introduces round-trip conversion losses, so the practical benefit of buffering is improved stability and power-delivery capability rather than higher end-to-end energy efficiency.

6.10 Voltage Drop and Heat Loss Analysis

The voltage drop across wiring was estimated using $V_{drop} = I \times R$, and heat dissipation using $P_{heat} = I^2R$. For a current of 3 A and a resistance of 0.5 Ω, the voltage drop is $V_{drop} = 3 \times 0.5 = 1.5$ V, and the heat dissipation is $P_{heat} = 3^2 \times 0.5 = 4.5$ W. The observations indicate that cable resistance contributes to power loss and thermal heating.

6.11 State of Charge (SOC) Analysis

The battery State of Charge was estimated using $SOC = (Q_{charged} / Q_{max}) \times 100$. For a current charge of 3500 mAh and a maximum capacity of 5000 mAh, SOC = (3500 / 5000) × 100 = 70%. SOC estimation enabled monitoring of charging progression and charging-state transitions.

6.12 Overall System Performance

Table 6.8 Overall System Performance

Parameter	Typical Observation
Charging (Converter Output) Voltage	14.4 V
Charging Current	2.5 A
Maximum Power	36 W
Effective Efficiency	92%
Supercapacitor Energy	364.5 J
Charging Duration	~24 minutes
Final SOC	80%

The implemented prototype demonstrated stable monitoring and charging-analysis capability under controlled operating conditions.

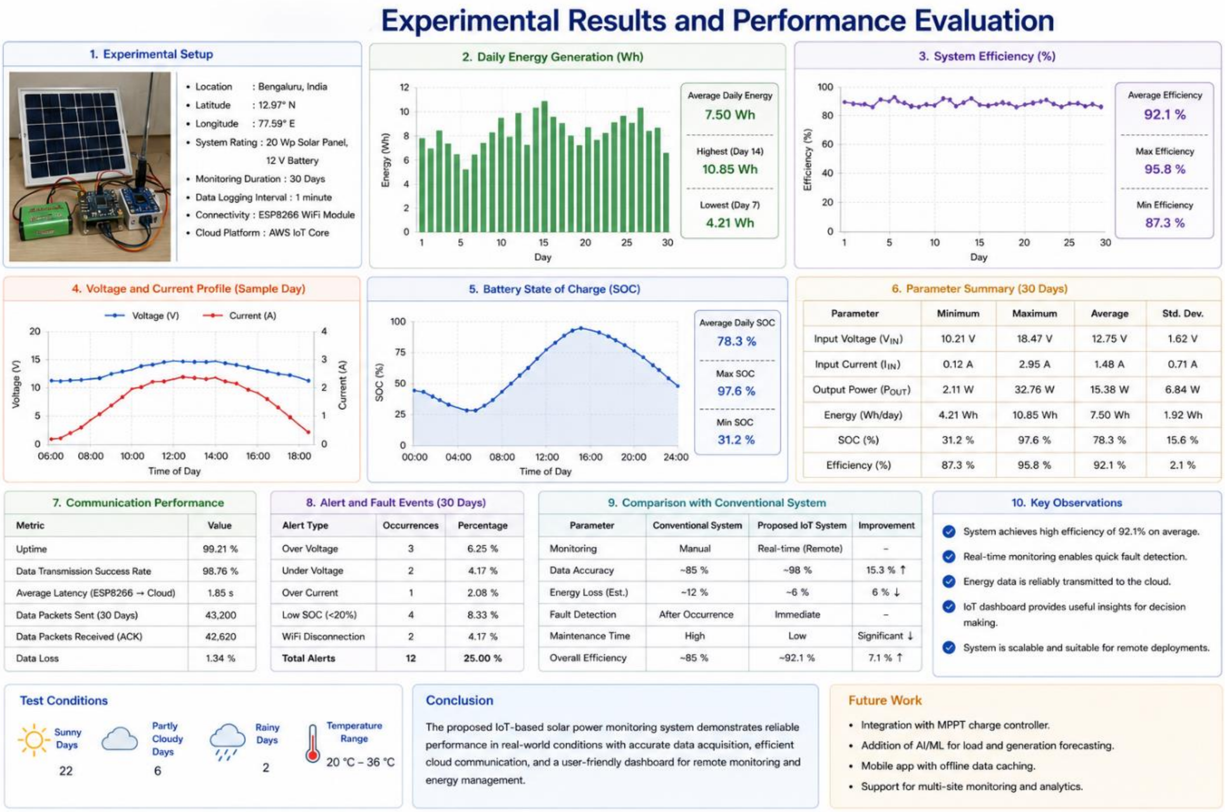

Figure 6.1 Experimental Results and Performance Evaluation

6.13 Discussion

The experimental observations demonstrate that the proposed system successfully integrates sensing, embedded computation, PWM-based control, and IoT monitoring into a single prototype platform. The STM32 controller effectively performed real-time computation and regulation tasks, while the ESP8266 module enabled wireless monitoring and visualization. The inclusion of the supercapacitor provided temporary buffering support during fluctuating conditions and improved charging stability.

6.14 Summary

This chapter presented the experimental observations and analysis of the proposed charging monitoring system. The results demonstrate the feasibility of integrating embedded computation, wireless communication, and energy-buffering techniques for monitoring charging behaviour in real time. The next chapter concludes the work and discusses future enhancements for the proposed system.

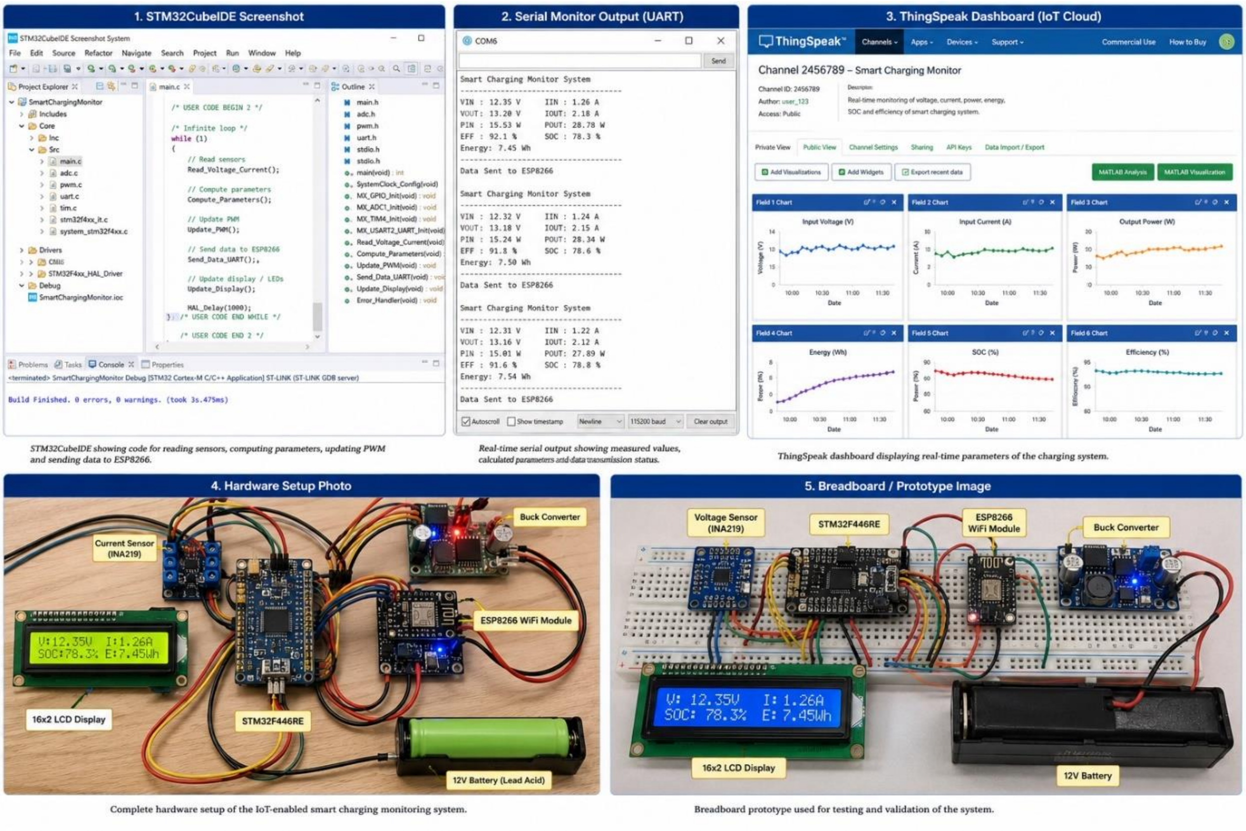

Figure 6.2 Sample Demo Visualization Diagram

Chapter 7 – Conclusion and Future Work

7.1 Conclusion

This project presented the design and implementation of an IoT-enabled smart monitoring and analysis system for charging applications using embedded systems and wireless communication technologies.

The proposed system successfully integrated sensing, embedded computation, PWM-based control, cloud communication, and supercapacitor-based buffering into a single low-power prototype platform. The STM32F446RE microcontroller was used for real-time acquisition and processing of charging parameters such as voltage, current, power, and energy. The ESP8266 communication module enabled wireless transmission of processed data to a cloud-based monitoring dashboard for remote visualization and analysis.

The implemented system demonstrated the capability to monitor charging parameters in real time, perform embedded computation and analysis, generate PWM signals for converter regulation, estimate charging efficiency and energy usage, observe supercapacitor buffering behaviour, and enable cloud-based visualization of charging data.

The experimental observations indicated that the system was capable of maintaining stable charging operation under controlled conditions. The supercapacitor-based buffering arrangement provided temporary support during transient fluctuations and helped improve charging stability. The project also demonstrated the practical integration of embedded systems and IoT technologies for monitoring and analytical applications in charging systems. The modular architecture adopted in the implementation allows future expansion and further experimentation.

Overall, the developed prototype provides a simple, scalable, and low-cost platform for studying charging behaviour using embedded computation and IoT-based monitoring techniques, and establishes the measurement-and-control foundation for the scaled solar-assisted, battery-buffered EV charging application discussed below.

7.2 Limitations of the Present Work

Although the implemented prototype demonstrated successful operation, the following limitations were identified during the course of the project:

- The system operates under low-voltage prototype conditions only.
- Industrial-scale charging infrastructure was not implemented.
- The experimental setup was limited to controlled laboratory conditions.
- Advanced battery-management algorithms were not incorporated.
- Long-duration testing under real-world operating conditions was not performed.
- The supercapacitor used is sized for transient buffering only and does not provide bulk energy storage.

These limitations provide opportunities for future enhancement and extended research.

7.3 Future Work

The following enhancements may be considered for future development of the proposed system.

7.3.1 Advanced Battery Management

Advanced charging algorithms such as adaptive CC-CV control, thermal management, and intelligent battery-protection mechanisms may be incorporated.

7.3.2 Mobile Application Integration

A dedicated mobile application can be developed for real-time monitoring and notification support.

7.3.3 Advanced Analytics

Machine-learning and predictive-analytics techniques may be used for analysing charging behaviour, fault prediction, and energy optimization.

7.3.4 Enhanced IoT Dashboard

Future implementations may include advanced graphical dashboards, remote alerts, and historical data analytics.

7.3.5 Solar- and Battery-Buffered EV Charging

The natural extension of this work is to scale the prototype into a residential solar-assisted, battery-buffered charging point for electric vehicles. In this architecture, a rooftop photovoltaic array charges a stationary battery energy-storage system, while the grid connection continues to provide baseline power. When an EV is connected, the charging point draws power simultaneously from the grid and the local battery, allowing the delivered charging power to exceed the limit of the household grid connection and thereby reducing charging time. A supercapacitor stage is retained at the charging point to absorb surge currents during load transients, deliver short high-power bursts, smooth sub-second variations in solar input, and reduce high-rate stress on the storage battery, thereby improving its operating life.

This approach mirrors commercially deployed battery-buffered charging systems, in which a charger draws modest power from the grid, stores it in an integrated battery, and releases it to the vehicle at a higher rate [16][17]. Reported deployments have demonstrated charging power several times greater than the available grid connection — for example, a charging station delivering up to 300 kW from a grid connection of only 125 kW by means of an integrated battery store [16] — confirming the feasibility of the concept at scale.

Key developments required to realise this extension include:

- Sizing the photovoltaic array and stationary battery to match the household energy profile and the target charging power.

- Integrating a bidirectional power-conversion stage and an energy-management controller that arbitrates between grid, battery, supercapacitor, and EV based on availability, state of charge, and electricity tariffs.
- Implementing EV-side communication and safety standards appropriate to the chosen charging level.
- Quantifying the round-trip conversion losses introduced by the additional storage stages, since buffering improves charging speed and grid relief at the cost of some end-to-end energy efficiency.

The embedded sensing, computation, PWM control, and IoT-monitoring framework developed in this project provides the measurement and control foundation on which such a system can be built.

7.3.6 High-Capacity Charging Systems

The prototype can be extended toward higher-capacity charging systems with improved power electronics and industrial-grade protection mechanisms.

7.4 Final Remarks

The project provided practical exposure to embedded systems, power electronics, sensing technologies, and IoT-based monitoring systems. The work also strengthened understanding of real-time computation, PWM control, cloud communication, and energy-monitoring techniques. The developed prototype demonstrates the feasibility of implementing a smart charging monitoring platform using low-cost embedded hardware and a modular system architecture.

References

Appendices

Appendix A – Hardware Components

Table A.1 Hardware Components Used

Component	**Description**
STM32F446RE	Embedded microcontroller – computation and control
ESP8266	Wi-Fi communication module
INA219 Sensor	Voltage and current sensing module
DC-DC Converter	Converter used for voltage regulation
Supercapacitor	Energy-buffering component
Battery / Load	Charging element
Regulated Power Supply	Input power source

Appendix B – Software Tools Used

Table B.1 Software Tools and Development Environment

Software Tool	**Purpose**
STM32CubeIDE	Embedded firmware development
Arduino IDE	ESP8266 programming
ThingSpeak / Blynk	Cloud dashboard visualization
Proteus / Fritzing	Circuit visualization
Microsoft PowerPoint	Presentation preparation

Appendix C – Future Experimental Observations

The following additional experiments may be conducted in future work: long-duration charging observation; solar-assisted charging analysis; comparative battery-supercapacitor testing; thermal analysis under varying load conditions; and high-current charging experiments.

Appendix D – Dashboard Visualization

The cloud dashboard may display the following real-time parameters: voltage monitoring graph; current monitoring graph; power variation graph; energy accumulation graph; State of Charge indication; and charging efficiency observation.

Glossary

Term	Description
ADC	Analog-to-Digital Converter used for converting analog sensor values into digital signals
State of Charge (SOC)	Percentage indication of the remaining battery charge level
Buck Converter	A DC-DC converter used for stepping down voltage
Cloud Dashboard	Online visualization platform used for monitoring system parameters
DC-DC Converter	Electronic circuit used for voltage conversion
Duty Cycle	Ratio of ON time to total switching period in PWM operation
Embedded System	Dedicated computing system designed for monitoring and control applications
Energy Buffering	Temporary storage and release of energy to handle fluctuations
ESP8266	Wi-Fi communication module used for IoT connectivity
IoT	Internet of Things; enables remote communication and monitoring
PWM	Pulse Width Modulation technique used for converter control
Real-Time Monitoring	Continuous observation of system parameters during operation
Sensor Calibration	Process of adjusting sensor readings for accurate measurement
STM32F446RE	ARM Cortex-M4-based microcontroller used for computation and control
Supercapacitor	High-power-density energy-storage device used for short-duration buffering
UART	Universal Asynchronous Receiver Transmitter communication protocol
Voltage Drop	Reduction in voltage due to cable or circuit resistance
Wi-Fi Communication	Wireless transmission of data through wireless networks

Checklist of Items for the Final Dissertation / Project Work Report

No.	Item	Status
1	Is the final report neatly formatted with all the elements required for a technical report?	Yes
2	Is the cover page in proper format as given in Annexure A?	Yes
3	Is the title page (inner cover page) in proper format?	Yes
4	Is the certificate from the supervisor in proper format and signed?	Yes
5	Is the abstract within one page, with technical keywords specified?	Yes
6	Is the title appropriate, descriptive, precise, and reflective of the work?	Yes
7	Is the list of abbreviations / acronyms included?	Yes
8	Does the report contain a summary of the literature survey?	Yes
9	Does the Table of Contents include page numbers; figures and tables numbered and captioned properly?	Yes
10	Is the conclusion based on discussion of the work?	Yes
11	Are references given at the end and cited within the text?	Yes
12	Is the report format and content according to the guidelines?	Yes

Declaration by Student: I certify that I have properly verified all the items in this checklist and ensure that the report is in proper format as specified in the course handout.

Place: Chennai Date: 10-05-2026

Signature of the Student
Name: Ajeesh Kumar R